Supplementary material

Search Strategy

Key search terms

blood pressure (Pressure, Blood Diastolic Pressure Pressure, Diastolic Pulse Pressure Pressure, Pulse Systolic Pressure Pressure, Systolic Pressures, Systolic）

Trajectory (trajectories**)**

Ischemic stroke(Ischemic Strokes Ischemic Strokes Stroke, Ischemic Ischaemic Stroke Ischaemic Strokes Stroke, Ischaemic Cryptogenic Ischemic Stroke Cryptogenic Ischemic Strokes Ischemic Stroke, Cryptogenic Stroke, Cryptogenic Ischemic Cryptogenic Stroke Cryptogenic Strokes Stroke, Cryptogenic Cryptogenic Embolism Stroke Cryptogenic Embolism Strokes Embolism Stroke, Cryptogenic Stroke, Cryptogenic Embolism Wake-up Stroke Stroke, Wake-up Wake up Stroke Wake-up Strokes Acute Ischemic Stroke Acute Ischemic Strokes Ischemic Stroke, Acute Stroke, Acute Ischemic**)**

prognosis (outcomes outcome predict prediction)

Search strategy

(((Ischemic stroke[Title/Abstract]) OR ((((((((((((((((((((((((Ischemic Strokes[Title/Abstract]) OR (Ischemic Strokes[Title/Abstract])) OR (Stroke, Ischemic[Title/Abstract])) OR (Ischaemic Stroke[Title/Abstract])) OR (Ischaemic Strokes[Title/Abstract])) OR (Stroke, Ischaemic[Title/Abstract])) OR (Cryptogenic Ischemic Stroke[Title/Abstract])) OR (Cryptogenic Ischemic Strokes[Title/Abstract])) OR (Ischemic Stroke, Cryptogenic[Title/Abstract])) OR (Stroke, Cryptogenic Ischemic[Title/Abstract])) OR (Cryptogenic Stroke[Title/Abstract])) OR (Cryptogenic Strokes[Title/Abstract])) OR (Stroke, Cryptogenic[Title/Abstract])) OR (Cryptogenic Embolism Stroke[Title/Abstract])) OR (Cryptogenic Embolism Strokes[Title/Abstract])) OR (Embolism Stroke, Cryptogenic[Title/Abstract])) OR (Stroke, Cryptogenic Embolism[Title/Abstract])) OR (Wake-up Stroke[Title/Abstract])) OR (Stroke, Wake-up[Title/Abstract])) OR (Wake up Stroke[Title/Abstract])) OR (Wake-up Strokes[Title/Abstract])) OR (Acute Ischemic Stroke[Title/Abstract])) OR (Acute Ischemic Strokes[Title/Abstract])) OR (Ischemic Stroke, Acute[Title/Abstract])) OR (Stroke, Acute Ischemic[Title/Abstract]))) AND ((Blood pressure[Title/Abstract]) OR (((((((((Pressure, Blood[Title/Abstract]) OR (Diastolic Pressure[Title/Abstract])) OR (Pressure, Diastolic[Title/Abstract])) OR (Pulse Pressure[Title/Abstract])) OR (Pressure, Pulse[Title/Abstract])) OR (Systolic Pressure[Title/Abstract])) OR (Pressure, Systolic[Title/Abstract])) OR (Pressures, Systolic[Title/Abstract]))))) AND ((((((prognosis[Title/Abstract])) OR (outcomes[Title/Abstract])) OR (outcome[Title/Abstract])) OR (predict[Title/Abstract])) OR (prediction[Title/Abstract])) AND (((trajectory[Title/Abstract])) OR (trajectories[Title/Abstract]))

The searches were limited to March 2022.

Adjust the search strategy appropriately according to the search rules for all the databases searched, and retrieved the following:

10（Pubmed ）

22（Embase）

20（Web of science）

Table 1:Study quality assessment criteria

Intervention Assessed	Criterion	Letter
Background/ Methods		
Aims/ Hypotheses	Aims/ hypotheses of the study clearly stated in the Introduction/methods	A
Description of the study population	The characteristics of the participants are described in detail, for example, age, gender, comorbidities	B
Medical ethics and Consent	Informed consent taken from all participants, and the study has ethical approval from the relevant committee	C
Sample size calculation	Formal sample size calculation performed at the beginning of the study	D
Inclusion and exclusion Criteria	Inclusion and exclusion criteria are clearly defined	E
Statistical Association between dependent and independent variables	Methods for assessing outcome between blood pressure trajectory and outcome are clearly defined	F
Statistical validation of associations between predictors and outcome	Associations between blood pressure trajectory and outcome assessed for statistical significance	G
Methods used to measure blood pressure	Methods used to measure blood pressure measurements clearly stated	H
Timing of blood pressure measurements	Timing of blood pressure measurements clearly stated	I
Methods for establishing blood pressure trajectory	Establishing of blood pressure trajectory clearly explained	J
Results		
Specification of relevant patient characteristics	Patient characteristics described: age, gender, stroke severity, antihypertensive medication, thrombolytic therapy, mean blood pressure	K

Adjustment for confounding factors	Analyses adjusted appropriately to consider confounding factors	L
Main results	Main outcome presented in a graph or table	M
Discussion		
Consideration for alternative explanation in the discussion	The strength of the evidence for each studied outcome is discussed	N
Discussion of study Limitations	Limitations of the study are described and discussed	O
Guidelines for future research	Any suggestion for future research is made	P

Table 2:Quality assessment of included studies

Study	Criterion																
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	results
Petersen NH et al.2022	1	1	0	0	0	1	1	1	1	1	1	1	1	0	1	1	12
Fan K et al.2021	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	14
Xu J et al.2019	1	1	1	0	1	1	1	1	0	1	1	1	1	1	1	1	14
Changwei Li et al.2018	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	14
Jie Xu et al.2019	1	1	1	0	0	1	1	1	1	1	1	1	1	0	1	0	12
Kim BJ et al.2018	1	1	1	0	0	1	1	1	1	1	1	1	1	0	1	1	13
Zheng et al.2021	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	13

Table 3:Data extraction

	Study Design	study population	Number of case	BP monitoring method and Equipment	Intervals between, and frequency of BP measurements	BPT groups and name	Major Outcome Measures and Follow up point	Key findings summary
Petersen NH et al.2022	observational cohort study	Anterior LVO stroke patients with EVT .	1791	Noninvasive BP measurements	every 2 hours during the first 12 hours, every 4 hours until 24 hours, and every 6 hours until 72 hours after EVT.	Five SBP trajectories: low,moderate,moderate-to-high, high-to-moderate, High.	(mRS) :90days	Patients with high and high to moderate SBP trajectories had significantly greater odds of an unfavorable outcome
Fan K et al.2021	Prospective study	AIS with IVT using alteplase within 4.5 h after symptom onset	353	Electrocardiogrammonitorto measure BP with patients lying in the supine position.	every 15 min for 2 h from the start of alteplase therapy, then every 30 min for 6 h, and then every hour for 16 h	Five SBP groups: Low drop-low SBP group; Rapid drop-low SBP group;rapid drop-medium SBPgroup;rapid drop-high SBPgroup;continuous	mRS score at 90 days	the slow drop-low level SBP group represent a well-established unfavorable outcome risk factor and extremely high SBP the continuous

						fluctuation-very high SBP group.		fluctuation very high level SBP group, are equally associated with elevated unfavorable outcome risk
Xu J et al.2019	prospective cohort study	AIS(with admission SBP ≥160 mm Hg)	316	a semiautomatic upper arm BP monitor	twice daily by nurses during the first 7 days after admission	Three SBP groups: Sustained high SBP, moderate decrease, and rapid decrease.	(mRS) score at 3 months	The decreasing trend was greater in the favorable than in the poor outcome group. A decrease in BP in the acute phase is predictive of favorable outcomes in IS patients
Changwei Li et al.2018	Retrospective study	IS within 48 hours of symptom onset and with SBP between 140-220 mmHg	4016	using a standard mercury sphygmomanometer and 1 of 4 cuff sizes based on participants"arm circumferences	baseline, every 2 hours for the first 24 hours after randomization, every 4 hours during the second and third days, and every 8 hours thereafter until hospital discharge or death.	Five SBP trajectories: High systolicBP; high-to-moderate-low systolic BP;moderate-high systolic BP;moderate-low systolic BP;low systolic BP;	combination of death andmajor disability(MRS score3-5) at months 3 and 24	Patient with moderate-low systolic BP during acute ischemic stroke had a lower risk of adverse clinical outcomes.
Jie Xu et al.2019	prospective cohort study	IS patients within 24 h of onset	1095	in the nonparetic arm,in the supine position, with a semiautomatic upper-arm BP monitor	During hospita-lization, BP was measured twice daily (6AM-9AM in the morning, 6PM-9PM in the evening)	Five SBP trajectories; consistent high BP; rapid decline from high to low BP;Delayed decline from high to low BP;consistent low BP;elevation from low to high;	major disability or death at 3 months,defined as MRS score 3 −6.	C1 (n=90) had the highest risk of poor outcome , while C2 and C4 had the lowest risk. C2 and C4 had a significant reduction in poor outcome risk when compared to C1.
Kim BJ et al.2018	prospective study	IS within 24 hours of symptom onset and had ≥5 SBP measurements during the first 24 hours of hospitalization	8376	BP measurements were made during the routine clinical practice of AIS patient care.	9 time points for 24 hours starting on the time of arrival with 3 hour-intervals and had ≥5 SBP measurements during the first 24 hours of hospitalization	5 SBP trajectory groups: low , moderate, rapidly stabilized, acutely elevated, persistently high SBP .	composite of recurrent stroke, myocardial infarction, and all-cause death(12 ± 2 months)	The risk of having vascular events was increased in the acutely elevated and the persistently high SBP groups but not in the rapidly stabilized group, when compared with the moderate SBP group.
Zheng et al.2021	Retrospective study	IS within 48 hours of symptom onset and with a systolic BP between 140-220 mmHg	3479	using a standard mercury sphygmomanometer based on participant arm circumference.	baseline, 14 days or hospital discharge, 3 months, and 12 months after stroke	4 trajectories; high-stable, high-decreasing, low-increasing, and low-stable.	combination of death or major disability (mRS score of 3 – 6) during 12 to 24 months after stroke onset	Patients with a high-decreasing or low-stable systolic BP trajectory after discharge had a lower risk of adverse clinical outcomes among patients with ischemic stroke.